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Philosophy of Science and Technologies examination - Paper 2

In this document, I discuss the nature of space presented by Professor Giovanni Valente during the course of Philosophy of Science and Technologies, held at Politecnico di Milano during the academic year 2025/26.

The assignment pursued is the following : *The problem of space is an outstanding topic in philosophy of science. Explain what are the metaphysical, epistemological and physical questions that arise in its context. Discuss in detail the debate between substantivalism and relationism, by outlining the position of Newton, Leibniz and Mach: in particular, review the various arguments in favour and against the two conceptions of space. Finally, present your own view on the matter. Not only list them but go over them. Confine your own ideas.*

Requirement: 2000 words.

Word count after this line : 2078

Introduction

Space can be seen as the answers to the question “where...?”, events happen, and we have an inner sense of location, position, orientation of objects that we use every day to localize. Location is used to experience, to act and to form knowledge, however space seems to be an invisible entity, we cannot sense. Matter does have a location, always in our experiences.

Metaphysically, one wonders about the ontology, what space is made of, and if it is distinguishable from matter. Would space exist if no matter was around? Is it independent from matter, or is it just a relationship between different objects?

Epistemologically, one wonders how to get to know more about such an invisible and non-sensible entity, can we learn through experience or one must use reasoning because more powerful and can go further? The nature of science discussed in the first paper, seems to play a fundamental role here, how should we get information and derive laws about space without observing it?

Physically, the question is about interaction between space and matter. Is the behavior of space influenced by matter, is the behavior of matter influenced by space? Can we describe them independently?

For example, in Einstein Relativity theories matter is “bending” space and this deformation is influencing other matter. Even though this results will arrive only in the 1900s, they help to understand how what we experience everyday may not be fully correct, and reasoning is needed to get to “better” knowledge.

Geometry

Let *geometry* indicate that branch of mathematics and reasoning concerned with properties of space such as the distance, shape, size, and relative position of figures. Nowadays, different geometries have been developed and studied; spaces can be described in different ways by different geometries.

Euclidean geometry

Euclidean geometry is the first geometry developed in history, and it was considered the only one for many centuries. It is associated with Euclide’s work *Elements*, where the Greek philosopher developed the theory from five axioms. For the sake of the length of this paper, I will omit here the assumptions and main results. The whole geometric theory is derived through proving by deduction the theorems.

Relationism

Relationism current of thoughts describe space as only given by relationships among bodies (constituted of matter). The concept of position of a body, only makes sense with respect to other matter in the surrounding. Therefore motion is also **always** relative, objects are in motion or at rest only with respect to some other objects, which are said to be taken as “reference frames”.

Motion and space are described through a *democracy* or *equality of all reference frames*. Talking about an absolute space is wrong, talking about a “true motion” is wrong because they do not exist.

Example : let’s take four cars named A,B , C and D. A,B travelling close to each other in the same direction, while C and D are parked near the road. If A and B maintain the same distance from each other, i.e. they travel at the same velocity (intended as a vectorial entity), then B is resting according to A’s reference frame and vice versa, while C and D are moving as all the world around. This behavior constitutes an argument in support of relationism.

However, relationism seems to crash when considering accelerated reference frames. Isaac Newton gives an alternative.

Absolutism – Newton

Space

Isaac Newton’s view of space is *absolutist*, that is there exist an absolute space according to which there exist a “true motion” and a “true position” of an object. The postulate is that there exist an absolute space and it is

described by Euclidean geometry. Any object has a “true position” and a “true velocity” and “true acceleration”. There exist a *preferred reference frame*.

Motion

Newton formulated three laws of motion, synthetically :

- (I) the principle of inertia
- (II) $F = ma$
- (III) “To every action (force F) there is always an equal (in intensity) and opposite force ($-F$).

The first principle of inertia was already described with Descartes laws (relationism). The second law of motion gives a quantification of the acceleration with respect to the force applied and the mass of the object, the more the force and the less the mass, the more the acceleration implied.

Results:

Newton’s whole idea is said to form an *inference to the best explanation*, that means even though his hypothesis may be not-observable (the absolute space), they are the ones best ones at explaining some observed effect. Indeed, postulating that there exist one absolute motion, it is possible to describe inertial effects.

Example: if an object is moving in a circular motion with constant speed, this motion is properly described by means of a *centripetal force*. If the reference frame is taken to be rotating together with the object and with origin at the center, and if such centripetal force is removed, the object is observed to move on the line passing through the object and the center of rotation, directed outwards. From such reference frame, one observes a force pulling the object away. This is the same phenomena as when in a car, during a turn, one experiences an outward force. This force due to the principle of inertia, and not by actions of some “real” forces, they are defined as *fictitious forces*. A fictitious force F is recognizable because it does not have it’s counter-force $-F$ as described in the third law of motion. This is not a real force, but if one fixes the absolute space then the centrifugal force can be described by means of the Inertia principle.

Moreover, absolutism can describe the “Bucket experiment”. Let a bucket full of water at half of his height, let it rotate (through a rope which was previously round up and then released), the observation describes water to start spinning with the bucket and, more importantly, the surface of the water goes *concave*. This phenomenon cannot be described by the previous and simple relationist view, if instead of the bucket one observer had to rotate around the bucket (so that relative motion between the bucket and the observer is the same), then the water will stay plain and not go concave! To get to this conclusion, one must assume that there is no suitable interaction between any of the surroundings (i.e. the earth) and the bucket of water;

Criticism:

Newton does not provide a proper explanation of how to recognize the absolute space. If a fictitious force is observed, it can be described by an infinity of reference frames, when all in constant linear motion with each other.

Relationism – Leibniz

Leibniz approached the problem with *rationalism*, in the meaning that experience can always be fallacious and knowledge should come from pure reasoning (geometry). He supported the idea that a theory to be true must not be supported by experimental evidences but by pure logic. His thought is based on one axiom:

There ought to be some sufficient reason why things should be in this way – Principle of sufficient reason (PSR)

Developing two thought experiments, according to Leibniz, relationist view must be the correct one.

1. *Static shift* : imagine there exist an absolute space, which could be described by Euclidean geometry. Consider the universe A and the universe B, where the relationships between objects inside universe A are exactly the same as the relationships between objects inside universe B (same relative position,

relative velocities and interactions, forces thus accelerations). Let A and B be at absolute rest. Let B be located in a different region of the absolute space, i.e. looking at universe B is the same as looking at universe A if statically shifted inside the absolute space. The two universes differ only from the location in the absolute space, however they would describe the same exact reality.

2. *Dynamic shift* : imagine the same situation, there exist an absolute space, which could be described by Euclidean geometry. Consider two universes as before A and B, but this time in absolute motion with two velocities v_a and v_b ; where the universe X has velocity v_x if all of his inner elements have velocity v_x . Let v_a and v_b differ by an amount constant in time.

Thus, Leibniz uses the Principle of identity of indiscernibles (PII):

To suppose two things indiscernible is to suppose the same thing under two names

Being that there exist no sufficient reasoning for which one configuration is more correct then the other, **and** being that there is no observable difference between A and B, **then** they must be the same. That is, any velocity and acceleration measured is actually relative, because absolute motion cannot be detected.

Criticism :

There is no real predictive power gained by Leibniz, and no physical application. At the same time, the bucket experiment is not properly explained if not by “internal forces in the water”. A relationist approach that will give a “better” explanation comes with Mach.

Relationism – Mach

Method :

Ernst Mach proposes a different view of relationism. He approaches sciences with an *anti-metaphysical censorship*, that is the belief that any metaphysical hypothesis cannot be accepted in a scientific reasoning. Thus, absolute space **cannot** be considered a true entity and cannot be used to describe reality. This rejection comes from a belief that physical and psychological phenomena coincide, because knowledge is a consequence of biological and cultural evolution of senses and minds, then science must adapt under the principle of experience:

science must found his roots in sensible experience.

This rejects Leibniz approach, but still lands to a relationist perspective, a different one.

Space and motion:

Mach's states that position and motion are relative, not only to its surroundings but to the whole distribution of matter in the universe, to the “distant stars”. That means that if one had to repeat the bucket experiment rotating the whole universe around the bucket, the water would go concave. This interpretation differs from what Newton would have predicted, that is the water, remaining still with respect to the absolute space, would not be subjected to any centripetal force.

But why? Because absolute space is not sensible, it is not observable, thus there is no scientific reasoning for thinking it exists.

Results and criticism:

Mach gives a theoretical description to inertia, but without giving a factual way of proving it. One could if he could move the whole universe, or stop interaction between a small system and the rest of matter.

Considerations

In my opinion our senses are corrupted by evolution, and not adapted as Mach said. Evolution is *energy saving*, it optimizes the gains of a physiological traits with the energies necessary to maintain them. For example, we cannot perceive all the wavelengths of light, even though experimentally they are analogous to the ones we

see. That is, we cannot perceive all reality. Moreover, I consider experience fallacious when it comes to causal relationships.

However, I consider our minds are as corrupted as evolution. Evolution may have not preserved some logical reasonings that were useless for life on earth. Mathematics is known to describe what is true for all humans, but this may differ from what is true in the universe. Thus, logic may be as fallacious as experience because both derivations of our minds.

Therefore, I stick with Newton's absolutism because of his inference to best explanation. Mach seems to not have given a full description of how would inertia work with his mechanics. Why will the water rise in the bucket if the whole universe is rotating around? There is not a complete dynamical theory as Newton's one.

However, one has to take into account General relativity in nowadays discussions about space. General relativity describes spacetimes as a dynamical geometric structure which is deformed by matter and energy. The invariance of the speed of light requires observers to measure different distances and time intervals. At the same time, mass produces gravitational time dilation and curves the geometry of space. This theory has been tested and successfully predicted different phenomena (as the deflection of light by gravity during the solar eclipse in 1919), it seems to be the best description we have of space, and thus we should accept its relationist approach.